



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Applications Development

APPLICATIONS DEVELOPMENT FOUNDATIONS 1 ADF152S

SUBJECT GUIDE:	2020
COURSE CODE:	ADF152S
NQF LEVEL:	5
PREREQUISITE SUBJECTS:	PREREQUISITE SUBJECTS: None

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ORGANISATIONAL COMPONENT

1. INTRODUCTION

Welcome to the 1st year of your DIPICT: Applications Development. Applications Development Foundations 1 (ADF152S) is the first of three parts of the Applications Development pillar that form this qualification. It is also one of three major subjects offered in your first year of study. The aim of the subject is give students a good practical programming foundation using Java as a programming tool.

The purpose of this subject guide is to provide you with written guidelines as to what to expect of the university and what the university in turn expects of you, specifically with regards to this subject. The subject guide provides you with administrative details of the department as well as the contact details of the lecturer(s) teaching this subject. The general course information shows you the number of lectures, practical and tutorial periods each week, and provides information about additional time you will be expected to spend on course-related work. The responsibilities of students with regard to participation and attendance are stated and comprehensive assessment details for the course are given in this section, as well as details of the materials that will be used, e.g. text books. Furthermore, technical details of the purpose of the qualification, course entry requirements and NQF level, assessment criteria and moderation are provided. The exit level and critical cross-field outcomes for the course are also included. In the last section, the modules and specific outcomes are shown and the work schedule provides a detailed weekly work plan for the year.

2. GENERAL

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator	Mrs E Zietsman	Engineering 2.10	(021) 460 3486	zietsmane@cput.ac.za
Lecturers	Mrs K Swart	Engineering 2.10	(021) 460-1800	swartks@cput.ac.za
	Mr W Rothman	Engineering 2.12	(021) 460-XXX	rothmanw@cput.ac.za
	Mrs S Tom	Engineering 2.14	(021) 464-7201	toms@cput.ac.za
Domain Coordinator	Mr K Naidoo	Engineering 2.48	(021) 460-3348	naidook@cput.ac.za
Head of Department	Dr B. Kabaso	Engineering 2.30	(021) 460-3713	kabasob@cput.ac.za
Secretary	Mrs N. Allie	Engineering 2.1	(021) 460-3010	alien@cput.ac.za
Administrative Assistant	Mrs D. O'Brien	Engineering 2.49	(021) 460-3685	pillaydl@cput.ac.za

Support Technician	S. Salie	Engineering 1.4	(021) 460-4209	saliesh@cput.ac.za
Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za
Support Technician	I. Saliem	Engineering 1.28	(021) 460-3031	saliemi@cput.ac.za
Support Technician	C. Sindayigaya	Engineering 1.28	(021) 460-3031	sindayigayac@cput.ac.za

UNDER CONSTRUCTION

2.2 Class Times and Venues

As published on notice boards. The subject consists of four practical periods per week.

2.3 Learner Management System (LMS)

<https://myclassroom.cput.ac.za/>: Blackboard is used for general communications and announcements, posting of course material, assignments and announcements. It is the responsibility of the student to check Blackboard regularly.

3. STUDY MATERIALS

3.1 Prescribed Textbook

Notes, references to websites and textbooks are available on Blackboard.

3.2 Recommended Readings

Notes, references to websites and textbooks are available on Blackboard.

3.3 Websites

Notes, references to websites and textbooks are available on Blackboard.

4. MODULE CREDITS

This module has 0.167 credits on NQF level 5. Please refer to the Faculty prospectus on <http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus> for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 200 notional hours allocated to this module. These notional hours comprise of 30 hours (30%) for contact time and the remainder will be self-study and other tasks, assignments, and tests.

METHOD	% OF YOUR LEARNING TIME
Lectures	30%
Independent self-study of standard texts and references and specially prepared materials (study guides, books, journals, using online tutorials etc.)	20%
Independent self-study, time needed to prepare for assessment (e.g. tests/exams)	25%
Student learning groups (working with your peers for group assignments)	25%

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Kindly refer to CPUT's Policy Docs and Student Guidelines:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.2 Class attendance and participation

Kindly refer to page 18 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.3 Grievance procedures

Kindly refer to page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.4 Plagiarism

Kindly refer to page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

*NB: An amended version of the General Handbook will be emailed to you during the first term.

6. ASSESSMENT

6.1 Assessment policy and regulations

This subject has a variety of assessments. Assessments are a combination of written theory tests, assignments and online quizzes

Test queries: Last day for any queries regarding a test is one week after the test was discussed in class. Thereafter the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be given to your lecturer (via the Secretary) within one week after the test was written. You must also inform your lecturer on the day of the test that you are ill.

Due dates must be strictly adhered to. NO late submissions are allowed for any subject in this Diploma.

Assignments are administered via Blackboard.

6.2 Assessment Opportunities: Administration

There will be a sick test in June for the first semester formal assessments and a sick test in November for the second semester formal assessments.

ASSESSMENT BREAKDOWN AND WEIGHTS							
ASSESSMENT CRITERIA	1 ST TERM	2 ND TERM		3 RD TERM	4 TH TERM		TOTALS
Nature of assessment	Written Test	Written Test	Quizzes Programs	Written test	Written test	Quizzes Programs	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	Yes	Yes	
Total assessment weight for the term	10%	25%	10%	20%	25%	10%	100%
Will the assessment be moderated?	Yes	Yes	No	Yes	No	No	

STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

At the end of the course the student must be able to:

1. Write a small application using JAVA objects to solve a problem.

7.2 Subject Structure

7.2.1 Subject Outcomes

SUBJECT OUTCOMES	DESCRIPTION
SO1	Understand, code and use the basic building blocks of a program, namely sequence, selection and iteration.
SO2	Work with data as a collection in a program.
SO3	Understand, code and use the building blocks of object-orientation, namely classes, inheritance and polymorphism.
SO4	Design, code and test a small application.

7.2.2 Subject Breakdown Schedule

SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES		ACADEMIC SCHEDULE	ASSESSMENT DATES
SO1	THEME 1: Building blocks of a program			
	SPO1: Use an IDE to code, compile and execute a basic JAVA program		Term 1	
	SPO2: Using data and variables		Term 1	
	SPO3: Reading data from a keyboard and do calculations		Term 1	
	SPO4: Code and use If and switch statements		Term 1	
	SPO5: Code and use loops (for and switch)		Term 2	
SO2	THEME 2: Work with a collection of data			
	SPO1: Code and use arrays		Term 2	

SO3	THEME 3: Object-orientation			
	SPO1: Design, code and use a classes		Term 3	
	SP02: Understand and apply inheritance to classes		Term 3	
	SP03: Add objects to an array and use polymorphism to display objects		Term 3	
SO4	THEME 4: Building a small application			
	SPO1: Methods and parameters		Term 4	
	SP02: Build small several applications		Term 4	

7.3 Critical Cross-field Outcomes

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display that responsible decisions, using critical thinking, have been made.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical, and/or language skills in the modes of written and/or oral presentation.
CCFO6	Use science and technology effectively and critically, showing responsibility towards the environment and the health of others.
CCFO7	Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.
CCFO8	Contribute to the full personal development of each student and the social and economic development of the society at large, by making it the underlying intention of any programme of learning to create awareness in individuals about the importance of: i) reflecting on and exploring a variety of strategies to learn more effectively; ii) participating as responsible citizens in the life of local, national and global communities; and iii) being culturally and aesthetically sensitive across a range of social contexts. (RSASAQA, 2001:24)

8. STUDY UNITS/STUDY THEMES

8.1 Study Units

Refer to 7.2.2 Subject Breakdown Structure.

8.2 Assessment Opportunities and Assessment strategy

Refer to 6.2 Assessment Opportunities: Administration for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment.

8.3 Subject Outcomes and Associated Assessment Criteria

Refer to 7.2.2 Subject Breakdown Structure for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment.

8.4 Self-study Activities

Exercises, quizzes, notes and videos available on Blackboard.

UNDER CONSTRUCTION

9. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY
STUDENT ACKNOWLEDGEMENT
2020
DIPICTA: INFORMATION TECHNOLOGY
APPLICATIONS DEVELOPMENT FUNDAMENTALS I
ADF152S

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATIONS DEVELOPMENT FUNDAMENTALS I** and understand the information and my responsibilities as a student for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname:
 (Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature:

Date: